



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI SATHYA SAI FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

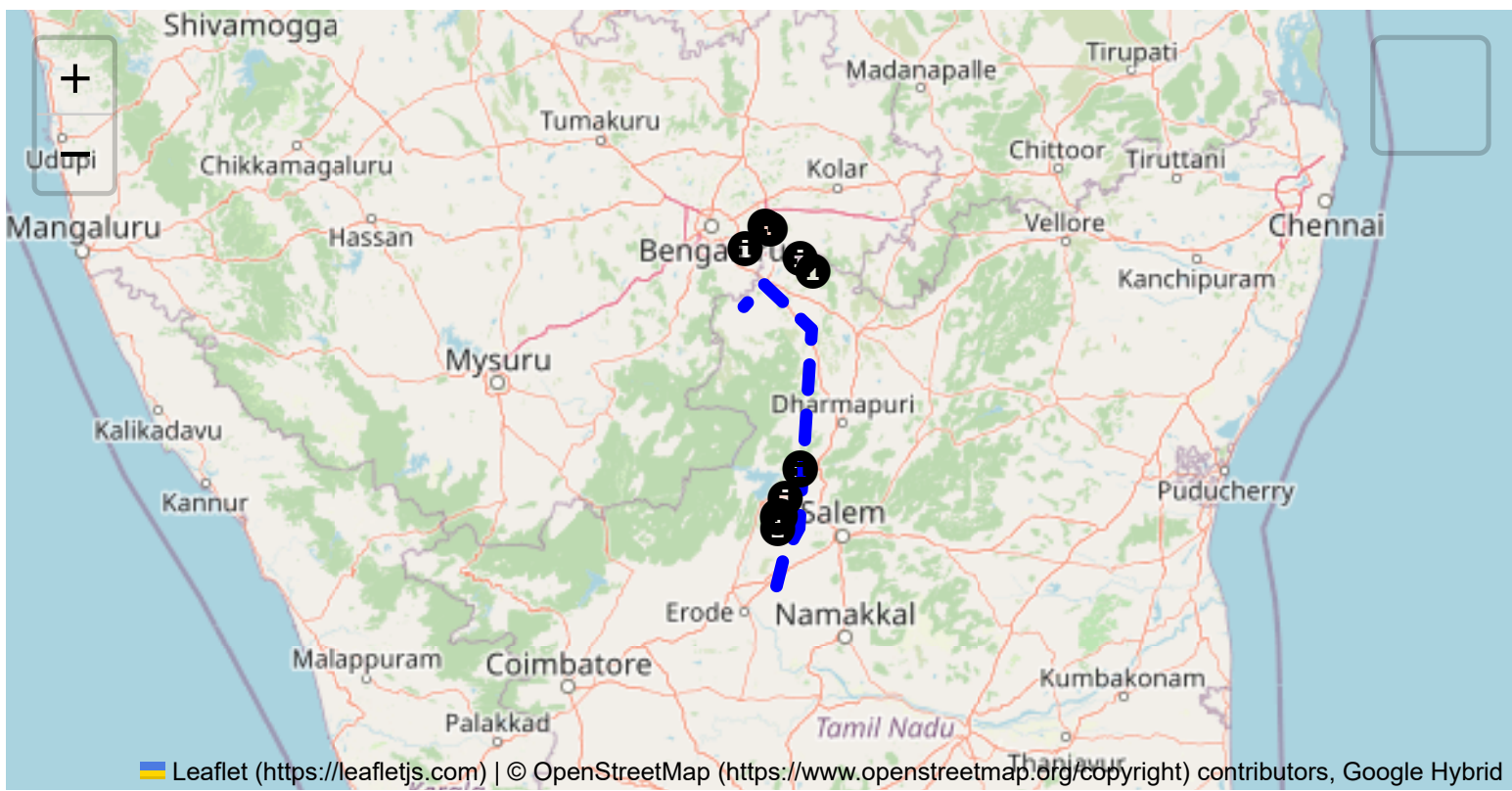
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

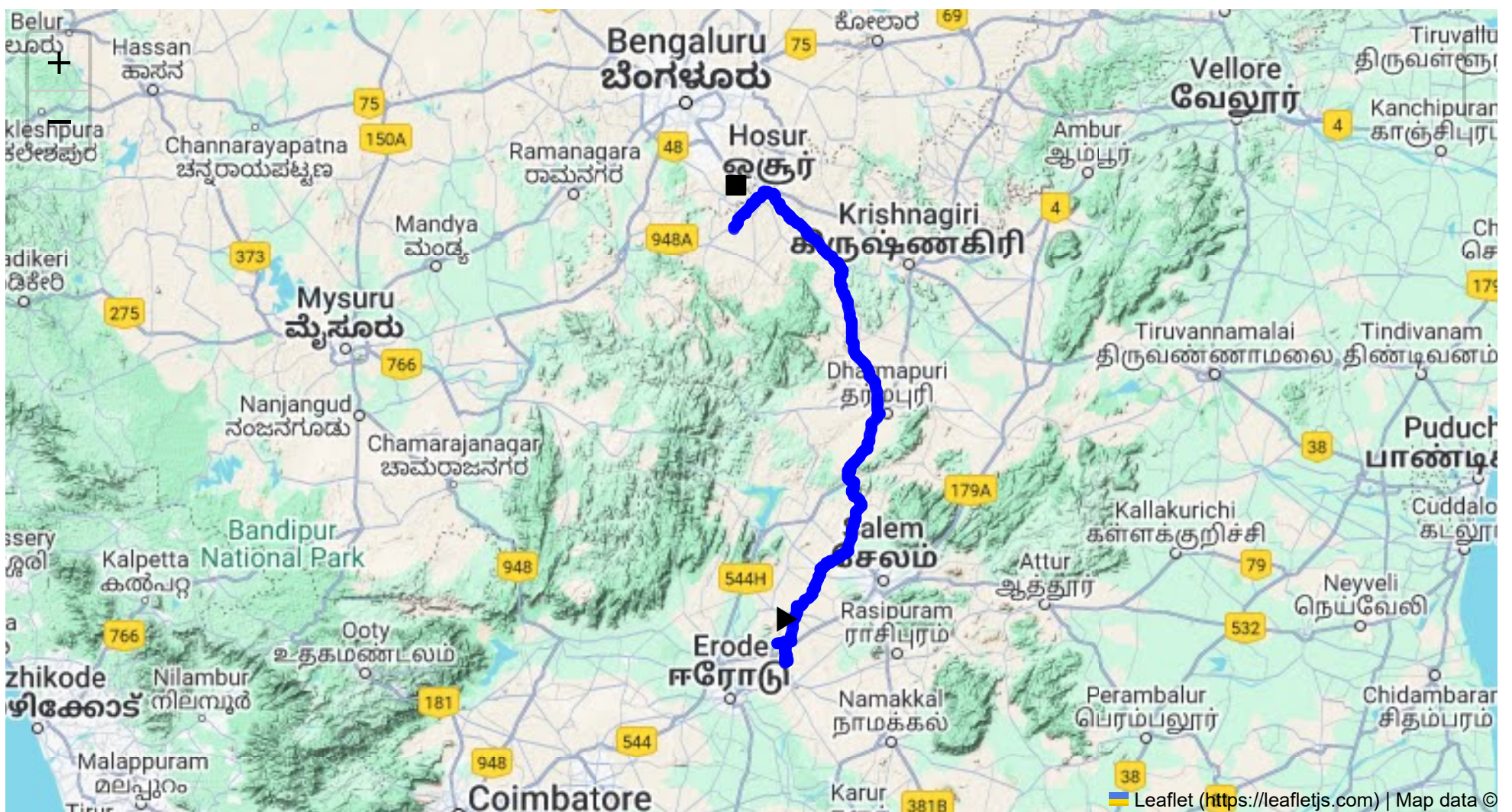
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 194.34 km
Estimated Duration: 4.3 hours
Adjusted Duration (Heavy Vehicle): 5.3 hours
Start: (11.4381, 77.8734)
End: (12.625277, 77.728577)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route stretches approximately 194.34 kilometers from Sangagiri to Hosur in Tamil Nadu, India. It traverses a diverse landscape, moving through rural areas, small towns, and some urbanized zones. This route involves state highways and local roads, with significant portions on SH 86. It begins at CVQF+23W,

Sangagiri, navigating through key localities like Idappadi, Sankari, Ayveli, Konganapuram, and Tharamangalam before reaching Hosur.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The weather in Tamil Nadu is usually tropical with hot summers, and the monsoon season can bring heavy rains between October and December. During the monsoon, roads can become slippery, and visibility might be reduced. Flooding can occasionally occur in low-lying areas, affecting road conditions.

3. Analysis of Traffic Patterns

Traffic can vary, with rush hours typically from 8:00-10:00 AM and 5:00-8:00 PM in more urbanized areas like near Hosur. Rural sections may see less congestion but be cautious of agricultural vehicles. SH 86 could experience moderate to heavy traffic due to its use as a key thoroughfare.

4. Assessment of Road Quality and Infrastructure

The state highways, including SH 86, are generally well-maintained but can have variable conditions, especially after heavy rains. Local roads might have narrow lanes, potholes, or less reliable surface quality. Regular maintenance issues can lead to sudden closures or diversions.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, routes via NH 44 or NH 544 could be considered, although they may require detours. These national highways offer better infrastructure and are typically cleared faster in emergencies but can increase travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transportation of hazardous materials requires adherence to Tamil Nadu's regulations, which often include securing permits and following prescribed routes during regulated hours, potentially avoiding dense urban areas during peak times.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific data may be limited, incidents involving heavy vehicles tend to occur on state highways. Majority involve collisions due to overloading or mechanical failures. Careful adherence to load and vehicle maintenance regulations is advised.

8. Environmental Considerations and Sensitive Areas

The route may pass near agricultural zones and smaller settlements where environmental sensitivity is higher. Proper containment and handling of hazardous materials to prevent spills or leaks are crucial.

9. Analysis of Communication Coverage

The route generally has good mobile network coverage. However, some rural or less developed segments may experience temporary communication gaps, especially in remote or low-lying areas.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times can vary widely. Urban areas like Hosur may offer quicker response due to proximity to healthcare and emergency services, while rural sections might see delays due to distance and local infrastructure.

11. Overall Summary of Risk Assessment

This route presents moderate risk for transporting hazardous materials. Key concerns involve weather-related road conditions, traffic congestion near urban centers, and road quality variability in rural areas. Adequate preparation, adherence to safety regulations, real-time monitoring of road and weather conditions, and alternative route planning can mitigate risks effectively. Always ensure communication devices are functional and information on local emergency services is accessible.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.71023, 77.79802	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	Medium	11.43995, 77.87431	30 KM/Hr
3	Turn	High	11.44107, 77.87528	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	Medium	11.47932, 77.87006	30 KM/Hr
10	Turn	High	11.48267, 77.87250	15 KM/Hr
11	Turn	High	11.49278, 77.86821	15 KM/Hr
12	Turn	Medium	11.49338, 77.86642	30 KM/Hr
13	Blind Spot	Blind Spot	11.49000, 77.85120	10 KM/Hr
14	Turn	High	11.49430, 77.88504	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	Medium	11.59143, 77.90623	30 KM/Hr
16	Turn	Medium	11.59297, 77.90746	30 KM/Hr
17	Turn	Medium	11.63836, 77.94819	30 KM/Hr
18	Turn	High	11.64176, 77.95188	15 KM/Hr
19	Turn	Medium	11.64660, 77.95114	30 KM/Hr
20	Turn	High	11.69562, 77.97113	15 KM/Hr
21	Turn	Medium	11.71394, 77.99572	30 KM/Hr
22	Turn	Medium	11.71739, 78.00032	30 KM/Hr
23	Turn	Medium	11.71984, 78.01216	30 KM/Hr
24	Turn	Medium	11.72770, 78.02297	30 KM/Hr
25	Turn	High	11.73787, 78.03528	15 KM/Hr
26	Turn	High	11.73655, 78.03891	15 KM/Hr
27	Turn	High	11.74265, 78.04710	15 KM/Hr
28	Turn	Medium	11.78017, 78.05790	30 KM/Hr
29	Turn	Medium	11.86727, 78.08317	30 KM/Hr
30	Turn	Medium	11.91282, 78.05854	30 KM/Hr
31	Turn	Medium	11.92500, 78.06476	30 KM/Hr
32	Turn	Medium	11.93385, 78.06370	30 KM/Hr
33	Turn	Medium	11.93757, 78.05319	30 KM/Hr
34	Turn	Medium	11.95570, 78.04715	30 KM/Hr
35	Turn	High	11.95870, 78.04990	15 KM/Hr
36	Turn	High	11.96031, 78.04945	15 KM/Hr
37	Turn	High	11.96291, 78.05131	15 KM/Hr
38	Turn	Medium	11.96453, 78.05092	30 KM/Hr
39	Turn	High	11.96562, 78.04928	15 KM/Hr
40	Turn	Medium	12.12050, 78.13284	30 KM/Hr
41	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
42	Turn	Medium	12.51395, 78.02897	30 KM/Hr
43	Turn	High	12.51632, 78.02571	15 KM/Hr
44	Turn	Medium	12.52916, 78.02035	30 KM/Hr
45	Turn	Medium	12.65342, 77.88784	30 KM/Hr
46	Turn	Medium	12.65573, 77.88629	30 KM/Hr
47	Turn	Medium	12.69040, 77.85486	30 KM/Hr
48	Turn	Medium	12.69448, 77.85412	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
49	Turn	Medium	12.71737, 77.83927	30 KM/Hr
50	Turn	High	12.71955, 77.83562	15 KM/Hr
51	Turn	Medium	12.71852, 77.83430	30 KM/Hr
52	Turn	High	12.71863, 77.83277	15 KM/Hr
53	Turn	Medium	12.72150, 77.83059	30 KM/Hr
54	Turn	High	12.72564, 77.81333	15 KM/Hr
55	Turn	Medium	12.71367, 77.80304	30 KM/Hr
56	Turn	High	12.71027, 77.79449	15 KM/Hr
57	Turn	Medium	12.66928, 77.76151	30 KM/Hr
58	Turn	Medium	12.62971, 77.72998	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
7	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
8	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
9	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
10	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
11	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
12	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
14	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
15	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
16	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
17	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
18	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
19	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
20	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
21	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
22	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
24	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
25	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
26	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
27	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
28	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
29	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
37	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
41	clinic	Narayana Hrudayalaya Superspecialty Clinic	12.7264081, 77.8242952	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
32	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
39	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
40	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium
42	school	Sishya school	12.7263093, 77.8145058	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71023, 77.79802



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44107, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47932, 77.87006



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48267, 77.87250



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49278, 77.86821



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49338, 77.86642



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49000, 77.85120



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49430, 77.88504



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59143, 77.90623



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59297, 77.90746



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63836, 77.94819

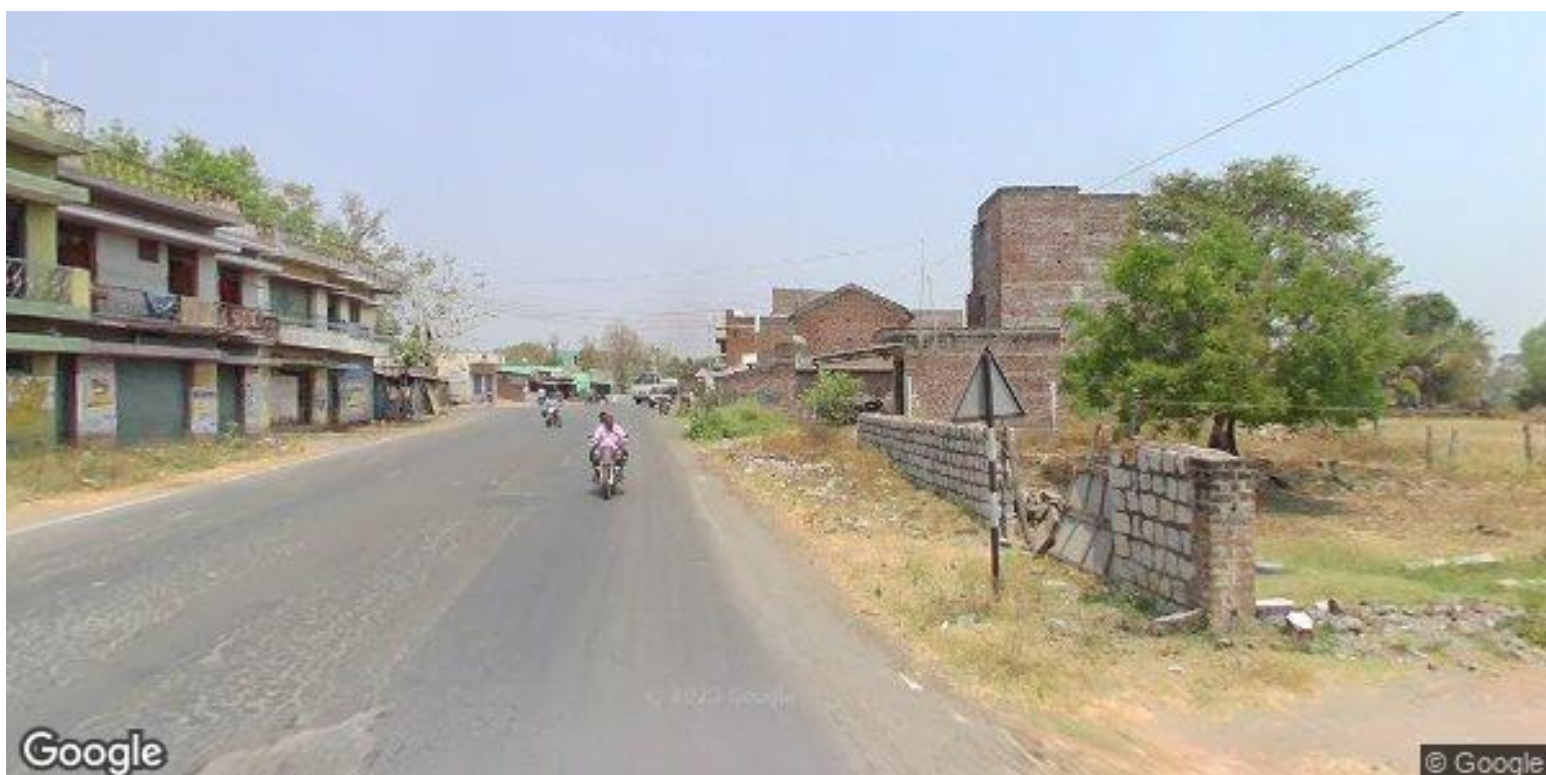


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64176, 77.95188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64660, 77.95114



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69562, 77.97113



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71739, 78.00032



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71984, 78.01216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72770, 78.02297



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73655, 78.03891



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74265, 78.04710



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78017, 78.05790



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86727, 78.08317



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91282, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92500, 78.06476



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93385, 78.06370



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93757, 78.05319



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95570, 78.04715



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95870, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96031, 78.04945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96291, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96453, 78.05092



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96562, 78.04928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12050, 78.13284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51381, 78.03208



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51395, 78.02897



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51632, 78.02571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52916, 78.02035



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.69040, 77.85486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.69448, 77.85412



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71737, 77.83927



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71955, 77.83562



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71852, 77.83430



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71863, 77.83277



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.72150, 77.83059



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.72564, 77.81333



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71367, 77.80304



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71027, 77.79449



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.66928, 77.76151
